

# KUNAL MUGALKHOD

## FULL-STACK GENAI DEVELOPER | AGENTIC AI & RAG

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### PROFESSIONAL SUMMARY

Full-stack developer and System Analyst with **13+ years of experience** across enterprise software, data engineering, machine learning, and clinical data systems. Builds GenAI applications from Angular interfaces and FastAPI services to **MCP integrations, RAG pipelines, and AWS EKS deployments**. Creator of Lightcode, a personal AI coding agent; hands-on experience with LLM gateways, authentication, observability, and performance optimization.

### CORE TECHNICAL SKILLS

**GenAI & Agents:** LLMs, LangChain, LangGraph, MCP, RAG, prompt engineering, Amazon Bedrock, Claude, OpenAI, LiteLLM

**Full-Stack:** Python, TypeScript, JavaScript, SQL, FastAPI, Flask, Angular, React, REST APIs

**Cloud & Delivery:** AWS EKS, Kubernetes, Docker, Helm, GitLab CI/CD, NGINX, Lambda, S3, API Gateway

**Data & ML:** PostgreSQL, DynamoDB, Weaviate, Pinecone, pandas, scikit-learn, TensorFlow, Spark, Hadoop

**Security & Quality:** Keycloak, OAuth 2.0, JWT, Langfuse, CloudWatch, Ragas, NeMo Guardrails, Pytest, JMeter

### FEATURED GITHUB PROJECT

#### Lightcode | Personal AI Coding Agent

[github.com/kmugalkhod/lightcode](https://github.com/kmugalkhod/lightcode) | TypeScript, React, Bun, Hono, Prisma, SQLite

- Built a coding agent with **terminal and localhost browser interfaces** sharing one local API server, agent engine, and persistent SQLite sessions; published as **@kmugalkhod/lightcode** on npm.
- Implemented permission-controlled file, shell, Git, web, and MCP tools, reusable skills, and runtime model switching across supported providers and OpenAI-compatible endpoints.
- Added **tiered context management** using token estimation, stale-output pruning, and conversation summarization while retaining complete session history in SQLite.

### PROFESSIONAL EXPERIENCE

#### Cybage Software Pvt. Ltd.

Aug 2021 - Present

System Analyst | Pune

Develop end-to-end GenAI applications, integrating enterprise systems with Angular, FastAPI, LLM orchestration, authentication, and cloud deployment. Partner with stakeholders on requirements, POCs, testing, and delivery.

#### Release Assistant | Agentic Release Intelligence

- Built **Jira and GitLab MCP servers** and agent workflows for release notes, environment-tag comparisons, sprint analytics, vulnerability reporting, and XRAY test-execution analysis.
- Reduced candidate tickets for commit analysis from **about 1,000 to 300** by filtering for tickets with commits before downstream processing.
- Integrated Bedrock/LiteLLM, Keycloak, and AWS EKS; delivered downloadable Word and CSV reports with supporting issue and repository links.

## PROFESSIONAL EXPERIENCE | CONTINUED

Cybage Software Pvt. Ltd. | System Analyst

### LiteLLM Gateway & SSO Portal

- Deployed a **LiteLLM gateway on AWS EKS** for Amazon Bedrock models and integrated Release Assistant and Claude Code through a common model-access layer.
- Built an Angular/FastAPI portal with **Keycloak SSO**, per-user and team virtual keys, budgets, usage visibility, and model-level cost tracking backed by PostgreSQL.
- Introduced a provider abstraction for Release Assistant to support direct Bedrock and LiteLLM-compatible APIs; configured Helm, ingress, and deployment automation.

### LTSA AI Assistant & LT Examination | Enterprise RAG

- Developed enterprise knowledge assistants with ingestion, chunking, embeddings, semantic and hybrid retrieval, metadata filtering, conversational context, and Cohere reranking.
- Integrated **Weaviate/Pinecone, Bedrock/OpenAI, and LangChain**; implemented NeMo Guardrails, response evaluation with Ragas, and Langfuse observability.

### API Catalog & OpenAPI Governance

- Built an API Catalog POC with frontend and backend services and a **reusable GitLab pipeline** to validate specifications and publish versioned API documentation to S3 on release tags.
- Developed and integrated **20 Spectral rules** for OpenAPI validation, with a dedicated pipeline validation stage to surface specification issues.

## ADDITIONAL ENGINEERING EXPERIENCE

**BPMN Generation & Land Title OCR:** Generated validated, Camunda-compatible BPMN 2.0 XML from business-process descriptions; built Amazon Textract extraction workflows for parcel, owner, and location data.

**Eyethena Healthcare:** Built Python REST APIs for data validation and PostgreSQL storage using AWS Lambda, Step Functions, API Gateway, IAM, and CloudWatch.

**Forecasting & Computer Vision:** Developed ARIMA, SARIMA, and Prophet forecasting workflows with MAE/RMSE evaluation; trained CNN and transfer-learning image classifiers using TensorFlow and Keras.

## EARLIER PROFESSIONAL EXPERIENCE

### Oracle Solution

Mar 2019 - Jul 2020

Senior Application Consultant | Pune

Performed data gathering, preprocessing, analysis, and basic model development.

### Veeva Clinical Research

Jul 2018 - Feb 2019

Associate Data Scientist | Ahmedabad

Built and deployed cloud software for pharmaceutical organizations to collect and process clinical-trial participant data, reducing manual work.

### Syneos Health

Feb 2013 - Jun 2018

Senior Clinical Data Programmer | Pune

Worked with Inform, Clintrial, and Central Designer; developed Base SAS reports and supported data processing and warehousing with Hadoop, Pig, and Hive.

## EDUCATION & CERTIFICATIONS

**Bachelor of Engineering (B.E.)** | Shivaji University | 2011

**C-DAC Professional Course** | 2012 | **Base SAS Certification**